An Exact Obstruction to the Local-Finiteness Premise in Fu's Theorem 2.8

Abstract

We give an explicit allowed polynomial system for which the regular-parameter set appearing in Theorem 2.8 of Y. Fu, *Towards Trans-Exponential O-minimal Expansion of the Real Ordered Field*, cannot possess the locally finite closed-simplex triangulation required by the argument. The example is the one-variable system $P(x, l_1, l_2, \epsilon, \delta) = x^2$. On a fixed identity-coordinate slice, the regular-parameter set is exactly $[-1, 1] \setminus \{0\}$, hence is not closed. A locally finite union of closed simplices in the full ambient parameter space is closed, giving an immediate contradiction. If the word "simplex" is instead interpreted as an open simplex interior, the membership conclusion used in Claim 2.9 does not follow from density and local finiteness. Thus the displayed construction route has a precise gap. This note does not claim a counterexample to o-minimality or a solution of the general trans-exponential problem.

1 The source premise

For a system P of the form used in Fu's equation (3), equations (4) and (6) define a parameter set G and the regular locus

$$R = \big\{(A, \eta, l, \epsilon, \delta) \in G \times [-1, 1]^{2n+2} : P \circ A - \eta \text{ has nonsingular zeros only}\big\}.$$

Theorem 2.8 assumes a triangulation $\{\Delta_\alpha\}$ of R with the following ambient local-finiteness property: every point of $G \times [-1, 1]^{2n+2}$ has a neighborhood intersecting only finitely many Δ_α. The proof then uses density of R to assert in Claim 2.9 that each point of the compact identity slice belongs to one of finitely many such simplices.

2 The explicit system

Take

$$n = 1, \qquad k = u = 0, \qquad P(x, l_1, l_2, \epsilon, \delta) = x^2.$$

Then $m = 3n + k + u + 2 = 5$. Fix arbitrary $(l, \epsilon, \delta) \in [-1, 1]^4$ and define

$$p_\eta = (I_5, \eta, l, \epsilon, \delta), \qquad -1 \le \eta \le 1.$$

Lemma 1. *For every $\eta \in [-1, 1]$, the pair (I_5, η) lies in Fu's set G.*

Proof. The zero set of $x^2 - \eta$ is empty for $\eta < 0$, consists of the single x-coordinate 0 for $\eta = 0$, and consists of the two x-coordinates $\pm\sqrt{\eta}$ for $\eta > 0$. Including the four auxiliary coordinates gives respectively the empty set, $\{0\} \times \mathbb{R}^4$, or $\{-\sqrt{\eta}, \sqrt{\eta}\} \times \mathbb{R}^4$. For any auxiliary point, restrict to a bounded neighborhood in those four coordinates. The inverse image under the auxiliary-coordinate projection is then empty, one bounded copy, or two bounded copies. This is exactly the local boundedness condition defining G. □

Proposition 1. *On the displayed identity-coordinate slice,*

$$\{\eta \in [-1,1] : p_\eta \in R\} = [-1,1] \setminus \{0\}.$$

Proof. For $\eta < 0$, the equation $x^2 - \eta = 0$ has no real zeros, so the assertion that all zeros are nonsingular is vacuously true. For $\eta > 0$, the only zeros are $\pm\sqrt{\eta}$ and the derivative $2x$ is nonzero at both. At $\eta = 0$, the zero $x = 0$ has derivative zero and is singular. These three cases prove the equality. □

3 Failure of ambient local finiteness

Theorem 1. *The regular locus R for the system above admits no triangulation by closed simplices that is locally finite in the full ambient space $G \times [-1,1]^4$.*

Proof. Assume that $R = \bigcup_\alpha \Delta_\alpha$, where each Δ_α is a closed simplex and the family is locally finite in the full ambient space. For $j \geq 1$, put $\eta_j = -1/j$. Then $p_{\eta_j} \in R$ and $p_{\eta_j} \to p_0$, while $p_0 \notin R$.

Choose an ambient neighborhood U of p_0 that meets only finitely many simplices, say $\Delta_1, \ldots, \Delta_N$. All sufficiently large p_{η_j} lie in U, so infinitely many of them belong to one fixed Δ_i. Because Δ_i is closed, their limit p_0 also lies in $\Delta_i \subset R$, a contradiction. □

Equivalently, a locally finite union of closed subsets is closed. The sequence above exhibits directly that R is not closed in the ambient space, so the first bullet of Theorem 2.8 cannot hold under the closed-simplex meaning used by its proof.

4 Why open simplex interiors do not repair the argument

One might instead call the relative interiors of simplices the members of a triangulation. Such pieces need not be closed, so the preceding topological contradiction would not apply verbatim. However, Fu's Claim 2.9 uses density and local finiteness to conclude that a limit point on the identity slice *belongs* to one of the finitely many simplices. For a decomposition by open simplex interiors, a sequence may converge to a proper face without the limit belonging to any of the interiors containing the sequence. The displayed points $p_{-1/j} \to p_0$ give exactly this failure mode. Thus either the simplices are closed, in which case the assumed ambient local finiteness is impossible, or they are open interiors, in which case Claim 2.9 requires a new argument.

5 Formal and independent verification

The accompanying Lean 4 development proves:

1. the unrestricted regular-parameter set for $x^2 - \eta$ is $\mathbb{R} \setminus \{0\}$;
2. the constrained slice is $[-1,1] \setminus \{0\}$;
3. this slice is not closed; and
4. it cannot be a locally finite union of closed pieces.

The development compiles with Lean 4.31.0, contains no proof holes, and passes the recorded axiom audit. A separate Reviewer checked the source definitions, the admissibility of the $P = x^2$ system, the mapping from the paper's parameter space to the formal slice, and the use of the closed-simplex interpretation in Claim 2.9.

6 Scope of the conclusion

The conclusion is a correction to the proof route, not a counterexample to the conditional conclusion of Theorem 2.8 and not a counterexample to o-minimality. The example proves that the stated triangulation premise is not available for all allowed systems and that the proof cannot obtain it merely from density of the regular locus. Consequently, the o-minimality construction depending on this step requires a replacement argument or a strengthened hypothesis.

Primary source

Yayi Fu, *Towards Trans-Exponential O-minimal Expansion of the Real Ordered Field*, arXiv:2604.03477v2. Relevant locations: equations (3), (4), and (6); Lemmas 2.5–2.7; Theorem 2.8; and Claim 2.9.